P1:- Solve the following problem with CPU Scheduling Algorithms.

Draw a table with 4 rows with 3 colums

first row

2nd row

Process	AT	BT
P_1	0	5
P_2	0	4
P_3	0	3

switch
if there is a context which in the CPU S.A then there is context switch time of 5 sec so each unit is considered as seconds.

Two Scheduling Algorithms

1) Preemptive 2) Non-Preemtive

Once CPU is assigned with anyone of the process

opp

3 S. Algorithms

1) → SRTF
Shortest Rem... Time first
(Preemption version of SJF)

2) PS

same

1) FCFS
first come f.

2) SJF
shortest Job first

3) PS
Priority scheduling

3) RR

round Robbing

for given problems

if 4 colums have we can apply all 6 algorithms (Process, AT, BT, Priority)

4 colums

if last colum is not there we can only 4 algorithms (Ps x)

(Process, AT, BT)

if Tq is not given we have to assume as 1

FCFS

Arrival of process

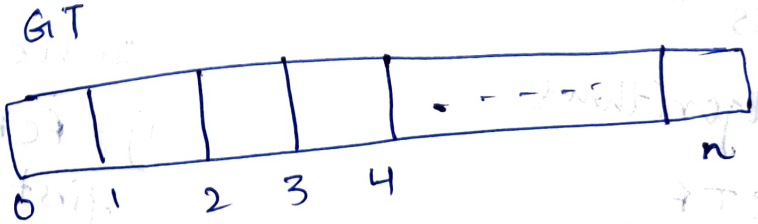
all three process are arrived at same time that is 0

Rq → Data structure shows process which are available for execution

GT

Rq

0: P1 P2 P3



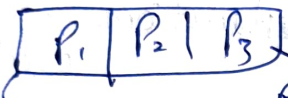
if all arrived at same then consider least bus time

also

0: 



because all 3 arrived at same time

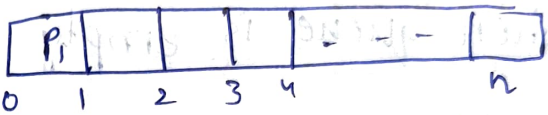
0: 

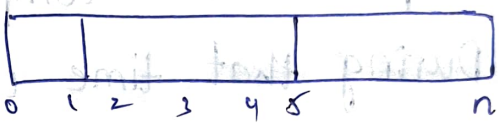
frontend

Rare end. (adding)

~~Add~~ removing

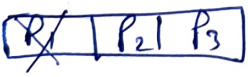
GT

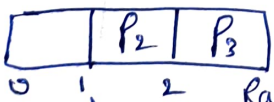




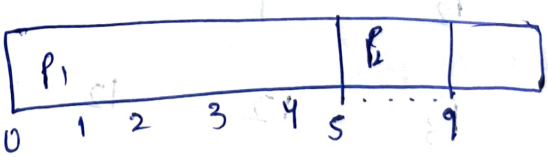
if we apply CPU to process it is remained with process until the proces execution is completed (Non-preemtive)

Amount of Time required for completion of process execution is called Bus time

5: 



GT



4 units required for P2

9: 

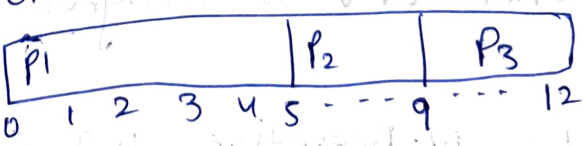
F=1 = +1 = 2 R=2

f=2=R=2 so

the last queue is having one element



GT



3 units to P3

f=R= equals

-1 ⇒ queue empty.

P_3 is completed at 12 second

P_3 is removed there

P_2 : 

F = 2 R = 2

now F = R = -1 

queue is empty

when queue is empty 2 situations might occur

1) All process are completed
2) During that time CPU

Using GT calculate CP completion Time, TAT, WT, & RT. ↑ Response time

Turn around time — waiting Time

	CT	TAT CT−AT	WT TAT-BT	RT AT−WT
P_1	5	5	0	0
P_2	9	9	5	5
P_3	12	12	9	9

When CPU is assigned for -- R.T.

for N. Preemtive WT & RT are same

CPU length

→ Starting position

Ending position of CPU - Starting position of CPU

$12 - 0 = 12$

Throughput :

No. of process executed / ~~within~~ length

$= 3/12 = 0.25$

per unit 0.25 no. of process are executed

maximizing Throughput & minimizing WT & RT

$$\text{Avg} = \frac{\text{Sum of TAT}}{\text{no. of process}} = 26/3$$

$$= \frac{\text{Sum of WT}}{\text{"}} = 14/3$$

* Is convation & starvation are applied to it or not
* Is context switching is happening or not

Which gives highest throughput & lowest Avg TAT & WT we can choose suitable algorithm

SRTF:-

Shortest Remaining Time first

Context switch :-

RQ (Shortest B.T)

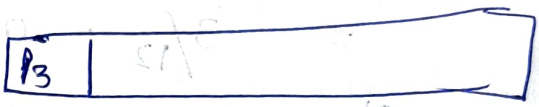
0 :- | P3 | P2 | P1 |

1 :- | P3 | P2 | P1 |

acc to ReT update RQ

1 :- | P3 | P2 | P1 |
no updation

GT



executed 1 unit out of 3 remaining = 2

ReT(P3) = BT − current execution Time (P3)

3 − 1 = 2

ReT(P2) = 4 − 0 = 4

ReT(P1) = 5 − 0 = 5

2 :- | P3 | P2 | P1 |
1 4 5

GT



ReT(P3) = 1

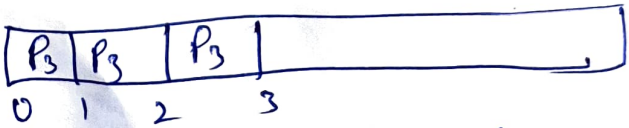
ReT(P2) = 4

ReT(P3) = 5

again select shortest R.T that is P3

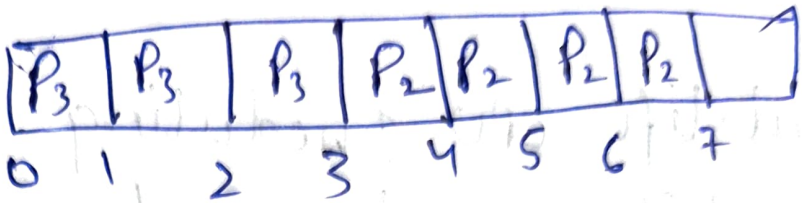
3 :- ~~| P3 | P2 | P1 |~~

GT



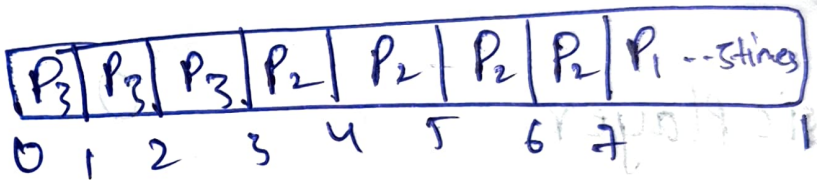
ReT(P3) = 0 ⇒ Process execu... is completed

GT



at 7 P_2 completed

GT



	CT	TAT CT-AT	WT TAT-BT	RT
P_1	12	12	7	
P_2	7	7	3	
P_3	3	3	0	

FCFS & SRTF Best becoz giving less WT

Q:- Real-life CPU scheduling scenario:

A mobile operating system has following tasks:

Task	AT	CPU Burst
Music Player	0	20
Touch Event	3	1
WhatsApp Notification	5	2
Camera Launch	6	4

Questions:-

1) Which scheduling algorithm is most suitable : FCFS, SJF, SRTF or RR ?
2) Draw the execution timeline for SRTF
3) Explain why the selected algorithm improve user experience.
4) Which task receives the fastest response time?

Ans

1) Suitable algorithm is SRTF because always checks the If a newly arrived task has a shorter remaining execution time than currently running task.

2)

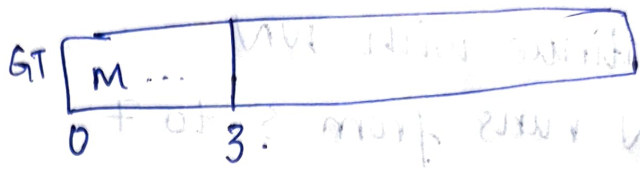
Execution timeline for SRTF is

M = music player

at Time 0:

AT = 3

~~BT = 1~~



Rel(MP) = 20 - 3 = 17

BurstTime - current execution Time

at Time 3:

T = Touch event arrives at 3.

BT = 1

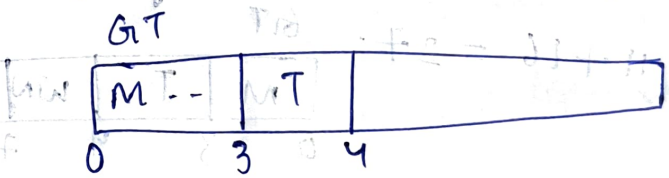
Rel(M) = 17

BT of TE (1) < Rel(M)

TE runs from 3 to 4

at Time 4:

Touch Event finishes



Rel(T) = 17 - 1 = 16.

at Time 5:

Whatsapp Notification

arrival time = 5

BT = 2

2 < 16

next AT = 6

at 6 WN has run for 1 unit

Rel(WN) = 1 . Camera launch arrives

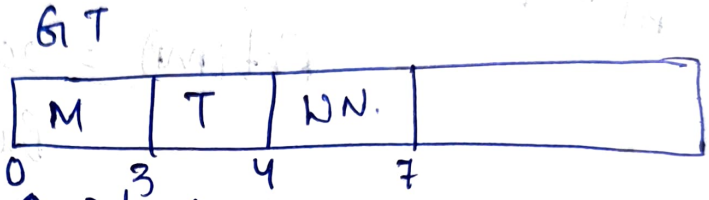
BT = 4

Now compare

Re(MP)(16) vs Re(WN)(1) vs CL(4)

Continue with WN

⇒ WN runs from 5 to 7



at time 7 (WN) finishes.

Now compare Re(M)16 vs CL(4)

CL is shorter so allocating CPU to CL

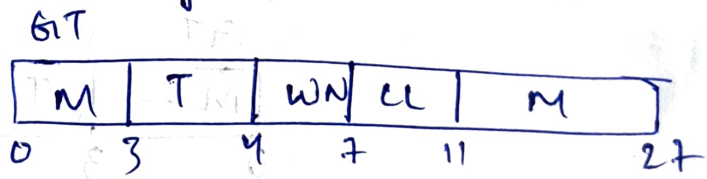
CL runs from 7 to 11.

at time 11 CL finishes.

now allocating CPU to MP

MP runs continuosly until completion

11 + 16 = 27.



0 3 4 5 7 11 26.

M | T | M | WN | CL | M

MP ⟶ TE ⟶ WN ⟶ CL ⟶ MP.

3) SRTF give a quick response to small task like touch events and notification so, the mobile beefs fast and responsive.

4) Touch

Task	AT	BT	first CPU	CT	TAT CT-AT	WT TAT-BT	RT
MP	0	20	0	27	27	7	0
TE	3	1	3	4	1	0	0
WN	5	2	5	7	2	0	0
CL	6	4	7	11	5	1	1

Music player, Touch Event & Whatsapp Notification share the fastest response time of 0.

3) Define following terms:-
Starvation, Convey effect, context switching Aging.

1) Starvation:- It occurs when a process waits indefinitely or for an excessively long time because other processes are repeatedly selected for execution. Starvation can occur in both preemptive & non-preemptive scheduling, depending on the algorithm.

Eg:- A low priority process may have keep waiting while high priority process are continuosly executed.

2) Convoy Effect:- It is an important problem associated with non-preemtive scheduling, especially FCFS. The long process acts like a convoy, causing many short processes to follow behind it

Eg:-

3) Context switching:-

The PCB stores the current execution state of a process, including the program counter, CPU registers, stack pointer, & other CPU related information. when the CPU switches from one process to another, the OS saves the state of currently running process in its PCB & loads the saved state of the next process from its PCB. This allows the interrupted process to resume execution from exactly where it was paused.

4) Aging:- It is a technique used to prevent starvation by gradually increasing the priority of a process that has been waiting for a long time.

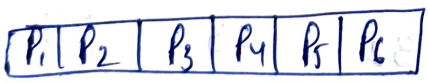
4)

Process	AT	BT	Priority
P_1	0	4	3
P_2	0	9	1
P_3	0	6	2
P_4	0	4	3
P_5	0	9	1
P_6	0	6	2
P_7	1	4	3
P_8	1	9	1
P_9	2	6	2

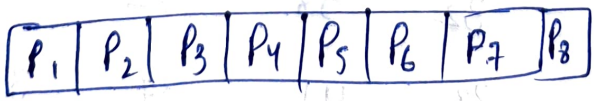
Apply all the CPU scheduling algorithms for insem ~~process~~

Sol: FCFS : First come first serve

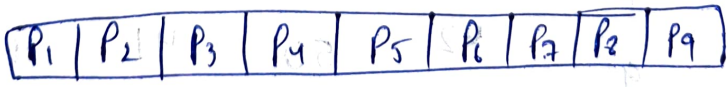
Rq 0:



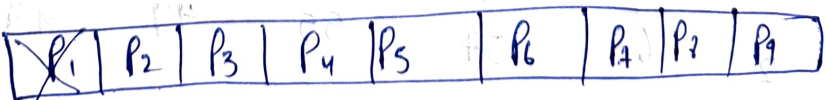
1:



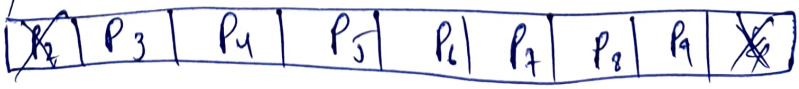
2:



4:



13:



19:



23 : | ~~P4~~ | P5 | P6 | P7 | P8 | P9 |

32 : | ~~P5~~ | P6 | P7 | P8 | P9 |

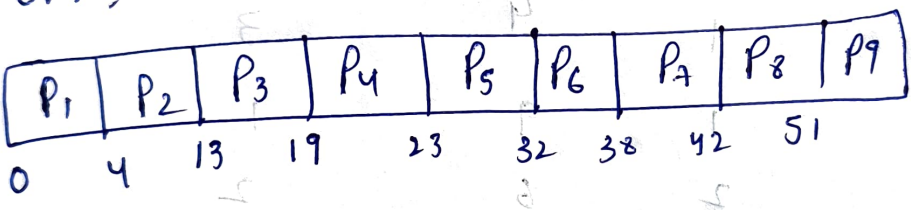
38 : | ~~P6~~ | P7 | P8 | P9 |

42 : | ~~P7~~ | P8 | P9 |

51 : | ~~P8~~ | P9 |

57 : | ~~P9~~ |

GT :-



Process	AT	BT	CT	TAT CT − AT	WT TAT − BT	RT WT − AT
P1	0	4	4	4	0	0
P2	0	9	13	13	4	4
P3	0	6	19	19	7	7
P4	0	4	23	23	19	19
P5	0	9	32	32	23	23
P6	0	6	38	38	32	32
P7	1	4	42	41	37	36
P8	1	9	51	50	41	40
P9	2	6	57	55	49	47